

highest probability of outperforming). The probability of outperforming \$0.30/share decreases at a slower rate as we execute more passively. This tells us that it is more beneficial to trade more passively than more aggressively when seeking to outperform a specified price.

Further Insight

Many times traders will specify the strategy in simpler terms. In some cases the strategies are well thought out and developed, but in other cases, they are simply instructions for the algorithms to follow and may or may not be in the best interest of the fund. These strategies include:

Volume Based. A volume-based strategy will instruct the algorithm to follow a specified trading rate such as 15 or 20% of the volume. At times, these trading rate values are consistent with the investment objective of the fund and thus achieve best execution, but in other cases they will not be consistent with the investment objective and will not achieve best execution regardless of the actual performance. The volume-based strategy could be in terms of trade rate α or percentage of volume *POV*.

Price Based. A price-based algorithm will instruct the algorithm to trade at a certain rate based on market prices. As prices become more or less favorable these algorithms will trade either faster or slower. These include algorithms such as ladder and step functions.

Hyper-Aggressive. A hyper-aggressive algorithm is one that executes as many shares as possible within a specified price level. As long as actual prices are more favorable than the specified price these algorithms will transact as aggressively as they can.

Passive/Dark Pool. The passive or dark pool algorithms are those that will transact primarily in dark pools and crossing networks. Usually these algorithms do not have a specified maximum trading rate such as is associated with the volume-based algorithms. They can participate with as much volume as possible provided they only transact in dark pools. Investors believe that if they are trading only in dark pools then they are minimizing their market impact cost and information leakage. While this is a widely held belief it is not correct. Market impact cost is caused by buying/selling imbalance. If you are on the side of the imbalance you will incur a higher cost. If you enter shares into a dark pool and your entire order is traded then there is a counterparty with at least as many shares as your order and possibly more shares, otherwise you would not have had your entire order executed. If you enter shares into a dark pool and only a portion of your order is executed then there is counterparty